Narrowband backed by mobile systems. Narrowband IoT (NB-IoT) networks are seeing a rapid rise since late 2015. The nudge was provided by the collaboration between Nokia, Ericsson and Intel Technologies to boost NB-LTE. Longer battery life and cheaper modules are the focus areas in the development of NB networks. Since it uses the backbone of the existing system, it provides for a better implementation on an economic scale.

This comes as a competing technology to Huawei's Narrowband Cellular IoT (NB-CIoT). Also known as NB-IoT, it looks to take over most of the smart city management, with applications ranging from smart water/gas metering, municipal light and waste management, livestock breeding to irrigation and environment monitoring. The biggest advantage of it is that its backing is based on mobile networks. This is, however, not based on LTE but instead on Direct Sequence Spread Spectrum (DSSS) modulation. It is expected to be simpler than NB-LTE-M with even cheaper chipsets.

The challenge with NB-IoT is that it does not fit in with LTE, hence, it calls for a separate side band with a different software setup, which could again increase deployment costs.

Lesser-known connectivity options. Sigfox is a viable alternative to LoRa, working in the same spectrum using binary phase-shift keying (BPSK). Even though it is bidirectional, communication via Sigfox tends to be better for uplink as compared to downlink. Downlink capacity is constrained due to limitations on receiving sensitivity at endpoint. This could well work for M2M communication, where we are concerned with the status of machines and status updates happen automatically.

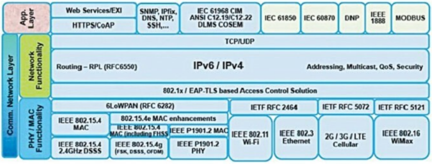

Fig. 2: IoT standards and protocols

Availability of Sigfox modules is not an issue anymore, as Sigfox is readily selling the endpoint technology to manufacturers. Availability of endpoints makes Sigfox extremely affordable. However, cost of a base station is high, making the system at par with LoRa network.

Thread is another technology that is being talked about. It targets to change the spotty and power-hungry nature of established connections. The mesh network created in a Thread connection looks to connect hundreds of devices around your home with minimum of batteries.

As per *threadgroup.org*, a simple AA battery could last for years. Aggarwal explains, "It spreads on the guarantee of an IP based work organising arrangement that is secure, solid, versatile and streamlined for low-power operation." Scalability of the network of up to 250+ devices and AES encryption to plug leaks in wireless connections add to the charm.

Weightless is a proposed wireless technology standard for exchanging data between a base station and thousands of machines around it using white space with high levels of security. White space refers to unused broadcasting frequencies in the wireless spectrum. This may not sound like much, but it becomes an important factor when you consider the capability to cover tens of kilometres in a single wireless hop. However, more realistic expectations from the system can be made after some solutions come to light.

Hardware and the open source community. There has been a lot of focus around the IoT with regards to open source hardware in the recent past. "The open source IoT framework allows you to utilise open-sourced implements to customise an IoT platform to suit your requirements," says Aggarwal. It provides a positive push for the initial start of product development.

Anand says, "However, due to mixed type of resource availability, product development is often left to be a subject of concern. We have a lot of IoT solutions, some of which are working towards making open source hardware applicable in real life."

A look at improved security in IoT systems

A lot of focus with the IoT is on Edge devices operating independently. Hence, we start with a secure booting process that verifies the authenticity and integrity of the software using cryptographically-generated digital signatures. Post setting up the device, access controls to remote users should be verified. If security is compromised, access control could ensure minimal control to unauthorised users—limited to entering access codes. Once the device connects to a network, it should verify itself with the network very much like Wi-Fi passwords. However, making it a non-cached system could be an improvement in this regard, in case any unauthorised entity gets this far.

The device also requires packet-inspection capability to control traffic for the device. This would be significantly different from the mainstream protocols that ensure no loss of data. And most important of all, security codes and hashes need to be updated regularly. This may be a huge task to undertake, but it could ensure ideal operation of the network.